WhatsApp AI Agent

Architecture Blueprint

A domain-agnostic system for building conversational AI agents that serve customers via WhatsApp, powered by LLMs.

Example: Cab Booking Service, Palakkad

System Architecture

End-to-end message flow from customer to AI agent and back

Customer
WhatsApp

→

Messaging Gateway
Twilio / Meta

→

App Server
Flask / Node

→

AI Engine
GPT-4o mini

→

Database
SQLite / PG

1. Customer sends a message on WhatsApp
2. Twilio/Meta forwards it as a webhook POST to your server
3. Server sends the message + context to the AI engine (LLM)
4. AI returns a structured response with intent + reply text
5. Server executes the action (booking, lookup, etc.) and replies via WhatsApp

AI Agent — The Brain

How the LLM processes each message and decides what to do

Persona & Guardrails

System prompt defines the agent's identity, personality, and strict boundaries

Customer Context

Name, phone, booking history, and recent conversation are injected

LLM Processing

GPT-4o mini (or any LLM) interprets intent and returns structured JSON

Action Execution

Server parses the JSON and executes: book, cancel, lookup, update DB

AI Response (JSON)

```
{
 "reply":"Booking confirmed!
  Pickup:Palakkad Fort,
  Drop:Malampuzha Dam,
  Driver:Suresh,Est:25 min,
  Fare:~₹200",

 "action":"create_booking",

 "action_data":{
  "from":"Palakkad Fort",
  "to":"Malampuzha Dam",
  "est_distance_km":10.5,
  "est_duration_min":25
 }
}
```

Domain Agnostic

Same architecture, any industry — just change the prompt and actions

Cab Booking

Book rides, assign drivers, calculate fare by duration

Healthcare

Appointment scheduling, reminders, report delivery

Restaurant

Table reservations, menu inquiries, order tracking

Real Estate

Property listings, site visit scheduling, price enquiries

E-Commerce

Order status, returns, product recommendations

Education

Class scheduling, fee payment, student queries

The core remains identical: WhatsApp + AI + Database. Only the system prompt, action handlers, and database schema change per domain.

Technology Stack

Production-ready, cost-effective, and scalable

Messaging Layer

Twilio WhatsApp API
Meta Cloud API (alternative)

Database

SQLite (dev) → PostgreSQL (prod)
All conversations logged

AI Engine

GPT-4o mini (default)
Swappable: Claude, Gemini, Llama

Hosting

Railway / Render / AWS
Auto-deploy from GitHub

App Server

Python Flask + Gunicorn
Alternative: Node.js / FastAPI

Monitoring

Admin API dashboard
Conversation logs + analytics

Estimated Monthly Cost: AI ~$2 | Twilio ~$5 | Hosting ~$5 | Total: ~$12/month for small-medium operations

Ready to Build?

This architecture adapts to any domain.
Let's discuss how it can work for your business.

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